

Soil macropores induced by plant root as a driver for vertical hydrological connectivity in Yellow River Delta

Lumeng Xie, Jiakai Liu, Yi Li, Peisheng Huang, Matt Hipsey, Mingxiang Zhang, Zhenming Zhang

Research Article

Soil macropores induced by plant root as a driver for vertical hydrological connectivity in Yellow River Delta

Lumeng Xie^{1,2}, Jiakai Liu^{1,2}, Yi Li^{1,2}, Peisheng Huang³, Matt Hipsey³, Mingxiang Zhang^{1,2} and Zhenming Zhang^{1,2,*}

¹School of Ecology and Nature Conservation, Beijing Forestry University, Beijing 100083, China, ²The Key Laboratory of Ecological Protection in the Yellow River Basin of National Forestry and Grassland Administration, Beijing 100083, China, ³Centre for Water and Spatial Science, School of Agriculture and Environment, The University of Western Australia, Perth 6009, WA, Australia

*Corresponding author. E-mail: zhenmingzhang@bjfu.edu.cn

Handling Editor: Zhonghua Ning

Received: 3 January 2024, **First Decision:** 18 February 2024, **Accepted:** 14 March 2024, **Online Publication:** 4 April 2024

Abstract

The protection and management of the wetland should consider the changes in hydrological connectivity (HC) caused by the structural modifications of the soil macropores. The main purpose of our work is to clarify and quantify the influence of the soil macropores volume on the vertical soil hydrodynamic process mechanically and statistically by taking the form of a case study in Yellow River Delta (YRD), and further reveal the vertical hydrological connectivity in this area. Based on X-ray computed tomography and constant head permeability test, the results showed a highly spatial heterogeneity of the soil structure in the YRD, hydraulic parameter (K_s) was negatively correlated with bulk density and positively with soil macropore volume, soil aeration and maximum water capacity. Using Hydrus 1-D software and the Green–Ampt model, we estimated the characteristics of the hydrodynamic process in the soil without macropores, then evaluated the effect of the soil macropore on soil hydrodynamic process by comparing the experimental results with the simulation results. We found that increasing soil microporosity improved the convenience of water movement, which would enhance the HC of the region. The results will further help to reveal the eco-hydrological process at a vertical scale in soil and provide a theoretical guide for wetland conservation and restoration.

Keywords: soil macropore, soil hydrodynamics, hydrological connectivity, wetland restoration, Yellow River Delta

植物根系诱导的土壤大孔隙驱动黄河三角洲垂向水文连通

摘要：植物根系诱导的土壤大孔隙及其水文功能是揭示生态-水文相互作用的关键因素，也是区域生态保护和恢复的理论和实践基础，基于土壤水动力学的小尺度水文连通研究能够促进研究者对水文连通内涵的认知。本研究以黄河三角洲为例，基于X射线计算机断层扫描(CT)技术和恒定水头渗透实验，量化了不同植被群落土壤大孔隙特征及垂向水动力过程；利用Hydrus 1-D软件和Green–Ampt模型，估算了该地区均质土壤的垂向水动力特征；通过与入渗实验结果的比较，阐明了土壤大孔隙对土壤垂向水动力过程的影响，进一步揭示该地区的垂向水文连通性。结果表明，黄河三角洲地区土壤结构具有高度的空间异质性，土壤大孔隙体积数刺槐群落>碱蓬群落>怪柳群落，同时土壤大孔隙体积随土壤深度的增加而降低；土壤饱和导水率(K_s)与土壤容重(BD)呈负相关，与土壤大孔隙体积(V)、土壤通气量(SA)和最大持水

量(MWC)呈正相关;土壤大孔隙体积能够显著提高土壤饱和导水率,促进水分快速运动。本研究所探讨的小尺度“植被-土壤-水文”协同作用机制,将为湿地垂向水文连通的恢复提供理论指导。

关键词:土壤大孔隙,土壤水动力,水文连通,湿地修复,黄河三角洲

INTRODUCTION

Hydrological connectivity (HC, hereafter) referring to the water-mediated transport of matter, energy and organisms between or within the hydrological factor cycles provides a framework to study the hydrological process and its ecological responses in different dimensions (Conte and Ferro 2019; Liu *et al.* 2021a; Zhou *et al.* 2023). Based on landscape structures such as runoff source network and stream network, horizontal and lateral HC can be analyzed (Xingyuan *et al.* 2023). For vertical dimension, root-induced changes in soil structure, such as pore clogging, cracking of macro-aggregates, amalgamation of micro-aggregates and generation of macropores can create or change continuous water flow path, thus controlling soil water movement and retention, playing a vital role in the hydrological and ecological functioning of watersheds (Liu *et al.* 2021; Lu *et al.* 2020). Water flow in macropores is a dominating process in structured loamy or clayey soils, whereas matrix flux is dominant in sandy soils (Jarvis *et al.* 2009). Understanding the response of HC to changes in soil physical structure is essential to ecosystem management and conservation (Jahanishakib *et al.* 2021; Wu *et al.* 2023; Zheng *et al.* 2022).

Compared to the bare area with a homogeneous soil structure, the growth of plant roots can modify the soil structure by promoting the formation of soil macropores, which may lead to spatial concentrations of water flow through unsaturated soil (Dai *et al.* 2020b; Kochiieru *et al.* 2020; Zhang *et al.* 2016). This has important implications for the rapid movement of solutes through soils and plays a critical role in small-scale HC (Banwart *et al.* 2018). Porosity, macropore number, volume and size distribution are important characteristics of soil macropores structure. Soil type, land use and vegetation type are important factors influencing those characteristics (Beven and Germann 1982; Hu *et al.* 2019, 2020; Zhang *et al.* 2019). The characteristics of macropores formed by biological factors and abiotic factors are different, which are embodied in size, shape, direction and function (Najm *et al.* 2010). It has been proved that the presence of continuous bio-pores created by the fauna or by roots rather than abiotic pores

significantly increases water permeability, solute transport efficiency and soil aeration (SA) (Bottinelli *et al.* 2017).

Evaluating vertical HC hinges on understanding the laws of vertical soil water movement and the impact of soil structure. Saturated hydraulic conductivity (K_s) is a hydraulic parameter of soil representing the ability of soil water movement through the soil profile when the infiltration capacity of the soil has been exceeded (Ilek *et al.* 2019; Mahmood and Mohammed 2020). Previous studies showed that the K_s is dependent on the number and size of active macropores because the macropores could accelerate flow rate of water in soil (Jarvis *et al.* 2009), and internal soil physical properties such as soil macropore structure, soil type and external environmental conditions such as land use and vegetation type were the controlling factors leading to great spatial heterogeneity of K_s (Horn *et al.* 2020; Huang and Shao 2019; Liu *et al.* 2021b).

Recent research on detailed soil macropore characteristics has heightened the need for X-ray computed tomography (CT), and the data are more accurate than in traditional methods such as dye-trace experiments and breakthrough curves (Liu *et al.* 2023). Using this technology, previous studies have combined CT-derived macropore features with static hydrological parameters, indicating that macro-porosity and path number are good indexes to predict K_s (Luo *et al.* 2010b), but there is a lack of research on quantification of the influence of the soil macropores on vertical soil hydrological process mechanically and statistically.

The positive coupling among vegetation, soil and hydrology is an important prerequisite for maintaining the health of wetland ecosystems (Ying *et al.* 2016). However, clayey saline soil, which has poor soil physical properties and low saturated hydraulic conductivity (K_s), is one of the main soil types in the Yellow River Delta (YRD) (Guo *et al.* 2023; Mao *et al.* 2016). Moreover, climate change and construction projects would influence the unique vegetation-soil-hydrology interactions negatively, which would lead to a decrease in K_s (Bonell *et al.* 2010). As a result, the limited freshwater resources cannot maintain the stability of the vegetation-soil-hydrology

system, affecting the HC within the wetlands and between wetland patches. This poses a threat to the hydrological cycle, the movement characteristics of soil water are closely related to the water security and eco-hydrological processes in the surrounding areas of the west coastline of the Pacific Ocean in China (Dai *et al.* 2020a).

In this article, we quantified the soil macropore structure by applying X-ray CT and the soil water movement process through the constant head permeability test and Hydrus 1-D software. We assumed that soil macropores promote vertical HC, the extent of which depends on macropore volume. This study aims to (i) understand the distribution of soil macropores in different plant communities, (ii) link the volume of soil macropores and other soil physical properties with K_s and (iii) explore the effect of soil macropores on the soil water movement process in the vertical dimension by comparing measured and estimated values. This study will elucidate the significance of plant root-induced soil macropores in vertical hydrological processes, providing scientific guidance for the ecological conservation and restoration of the YRD.

MATERIALS AND METHODS

Study site and soil samples

The study was conducted at the northern part of the modern Yellow River channel. This region has

a warm temperate to continental monsoon climate, the mean annual temperature is 12.1°C, and the mean annual evaporation is 1962 mm, which is 3.5 times the precipitation (Bai *et al.* 2019). The zonal soils are developed by the Yellow River alluvium, and fluvo-aquic soil and saline soil are the main soil types. The dominant plants in the region are *Suaeda glauca*, *Tamarix chinensis* and *Robinia pseudoacacia* (Fig. 1; Altieri and Cui 2015; Liu *et al.* 2020).

Three sample plots (1.0 m × 1.0 m) were randomly assigned in the *S. glauca*, (A1), *T. chinensis* (A2) and *R. pseudoacacia* (A3) communities. Firstly, we carefully removed grass and litter and collected soil samples from a depth of 0–60 cm using ring-cut for all 3 plant types. And then, a PVC cylinder (7 cm diameter, 10 cm height) with a beveled edge at its bottom was used to house the intact soil from the vertical profiles we excavated. The sampling depths were 0–10, 10–20, 20–40 and 40–60 cm, respectively. After sampling, the cylinders were wrapped in sponge and put into a box filled with wheat straw as shock-mounted elements to provide additional protection to the samples during transportation. We also took soil samples with a foil sampler near the study area to analyze soil properties.

We used Mastersizer 2000 (Malvern Panalytical Ltd, UK) to directly determine the soil particle size after full dispersion, following the American soil particle size standard. The soil water content θ_i was determined by TRIME-PICO TDR instruments (IMKO Co. Ltd, Germany), the soil bulk density

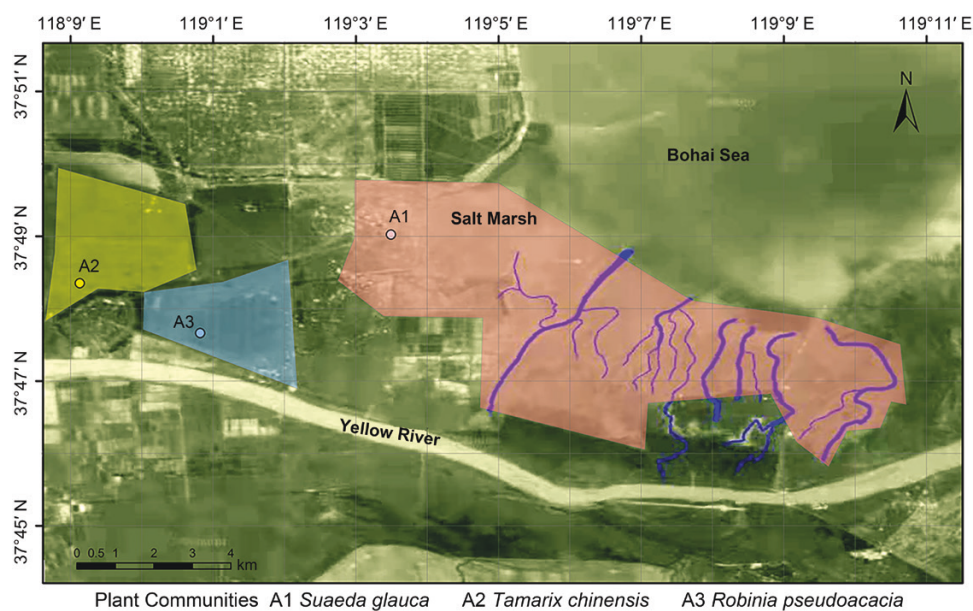


Figure 1: Location of the study area of three plant communities.

(BD) and the maximum water capacity (MWC, θ_m) was determined by the ring-cut method, and SA was calculated by the following formula:

$$BD = \frac{w_d}{V} \quad (1)$$

where w_d is the dry soil weight in ring-cut and V_r is the volume of ring-cut.

$$f = \frac{w_w - w_r - w_d}{V_r} \quad (2)$$

where f is the total porosity of soil, w_w is the weight of wet soil in ring-cut after 6 h of water absorption, and w_r is the weight of ring-cut.

$$SA = f - \theta_i \times 0.1 \quad (3)$$

where θ_i is the soil water content.

CT scanning and image analysis

An industrial CT equipment (Geoscan-100G, resolution $\leq 100 \mu\text{m}$) was applied to scan the undisturbed soil column. The image processing process is shown in Fig. 2. The volume of the macropores was calculated by the following formula:

$$V = A \times H \quad (4)$$

where V is the volume of the soil macropore (cm^3), A is the area of the soil macropore in a soil column

cross-section (cm^2) and H is the height of a soil column cross-section (cm).

Constant head permeability test

We performed the constant head permeability test to determine the K_s of the soil-column samples. We applied 1 g Vaseline on the place where the soil sample make contact with the pipe wall to prevent water from rapidly infiltrating through the gap. To observe the position of the wetting front clearly, the test was carried out with 1 g/l of Brilliant Blue solution, and we recorded the time (t) when the wetting front reached the bottom. The constant water head of 2 cm was controlled by a Mariotte bottle. When the outflow velocity of the undisturbed soil column became stable, a 10-ml measuring cylinder was used to record the volume of outflow every 1 min and then every 2, 5 and 10 min. K_s was calculated by Darcy's law:

$$K_s = \frac{QL}{AtH} \quad (5)$$

where K_s is the saturated hydraulic conductivity (cm/min), Q is the volume of permeated water (ml), L is the length of the soil column (cm), A is the cross-sectional area of the soil column (cm^2), t is the

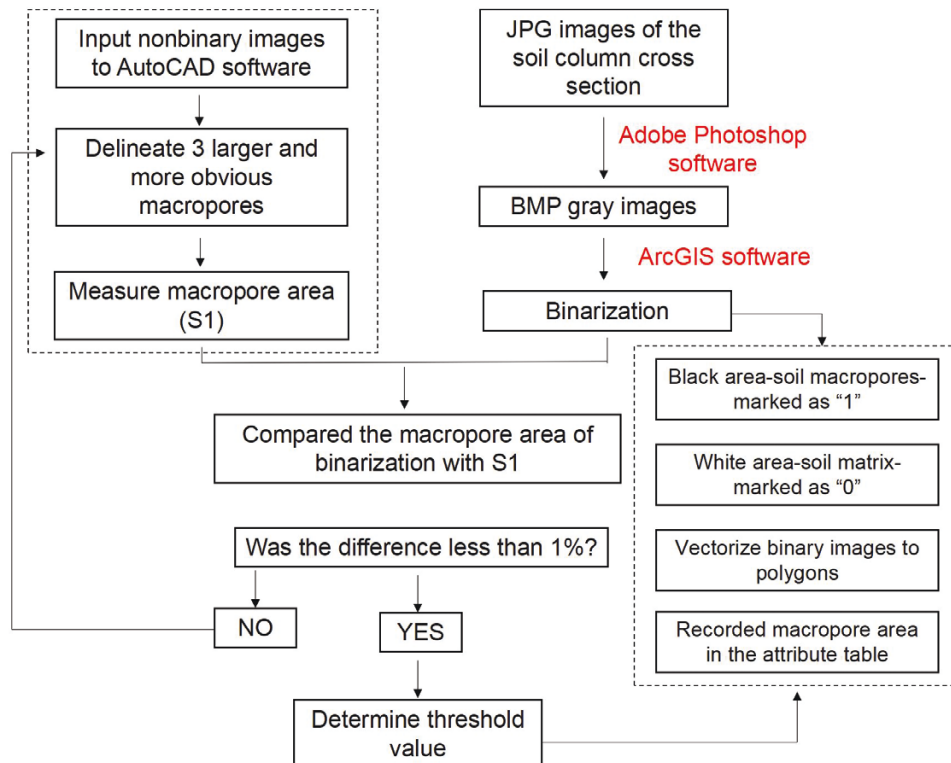


Figure 2: The flowchart for CT image processing.

permeation time of Q (min) and H is the height of the hydraulic head (cm).

Soil water dynamics calculation

We used the neural network prediction of Hydrus 1-D software to predict the saturated hydraulic conductivity (K'_s) according to pedotransfer functions in our study area. We input the soil textural class (American soil particle size standard) and BD information. K'_s was calculated, and the value represented the ability of soil water movement without soil macropores. Then we used the Green-Ampt model to estimate the time of water movement in the soil without macropores.

At the soil surface, the total head was H , as for $z(t)$, and at the wetting front, the total head was $-(h_c + z_f)$. Using Darcy's law, we obtained

$$q = v_t = -K'_s \frac{(h_c + z_f) - H}{z_f} \quad (6)$$

$$v_t = K'_s \frac{h_c + z_f + H}{z_f} \quad (7)$$

$$I = (\theta_s - \theta_i) z_f$$

$$i = \frac{dI}{dt} = (\theta_s - \theta_i) \frac{dz_f}{dt} \quad (8)$$

Using the water balance principle, we obtained

$$\frac{dz_f}{dt} = \frac{K'_s}{\theta_s - \theta_i} \times \frac{H + h_c + z_f}{z_f} \quad (9)$$

Integrating Equation (9), we obtained

$$\int_0^{z_f} \frac{z_f}{H + h_c + z_f} dz_f = \int_0^t \frac{K'_s}{\theta_s - \theta_i} dt \quad (10)$$

$$t = \frac{\theta_s - \theta_i}{K'_s} \left[z_f - (h_c + H) \ln \frac{z_f + h_c + H}{H + h_c} \right] \quad (11)$$

When $z_f > H + h_c$, omitting $h_c + H$, we applied the following equation to calculate the time (t_s) when the wetting front reached the bottom under the condition of the soil without macropores:

$$t_s = \frac{\theta_s - \theta_i}{K'_s} z_f \quad (12)$$

Statistical analysis

We calculated the soil macropore data using Excel. Pearson correlation analysis and principal component regression method were performed to analyze the correlation between K_s and the soil physical properties. The Durbin-Watson test was performed to test the regression model. Regression analysis was used to reveal the response of soil water movement to the soil macropores. We used R version 4.1.3 to conduct all the statistical analyses and to compile graphical outputs.

RESULTS

Characteristics of soil macropores

The macropore volume at different soil layers of the three plant communities is listed in Table 1. The microporosity variation with soil depth was significantly different between the 3 plant communities (Table 1). In A1, the highest volume was 15.81 cm³ at 10–20 cm soil layer depth, and the lowest volume was 2.95 cm³ at 40–60 cm soil layer depth. In A2 and A3, the macropore volume decreased with the increase in soil depth. However, we observed that in A3, the volume at 0–10 cm soil layer depth was 40.47 cm³, 50.6 times larger than the volume at 40–60 cm soil layer depth. All the samples indicated that the deeper the soil depth, the smaller the macropore volume and the volume in A2 was the lowest. This variation showed the spatial heterogeneity of the distribution of soil macropores.

Table 1: Macropore volume at different soil layers of the three plant communities

Soil depth (cm)	Macropore volume (cm ³)				
	0–10	10–20	20–40	40–60	Total
A1	5.92	15.81	4.59	2.95	29.27
A2	3.36	3.29	0.83	0.49	7.97
A3	40.47	12.55	1.33	0.80	55.15

A1: *Suaeda glauca* community; A2: *Tamarix chinensis* community; A3: *Robini apseudoacacia* community.

Factors controlling K_s variation and their relationship

Basic characteristics of K_s and soil properties

In our study area, the K_s values for all the samples ranged from 0.003 to 0.273 cm/min. The coefficient of variance was 1.26, which showed a highly variable trend. The mean K_s value was 0.013 cm/min in A1, which was smaller than that in A2 (0.084 cm/min) and A3 (0.124 cm/min), although the difference was not significant ($P > 0.05$). Soil physical properties are listed in Table 2, the values are higher in A3.

Construction of a multiple linear regression model between K_s and soil physical properties

The Pearson correlation coefficients between K_s and the factors obtained are displayed in Fig. 3a. K_s was significantly related to all the factors selected ($P < 0.01$). K_s was negatively correlated with BD and positively correlated with volume, SA and MWC. The regression coefficient table in stepwise regression indicated that multicollinearity exists in these factors (variance inflation factor > 10). So, we chose principal component regression analysis to set up the relation equation that combines K_s with other factors. Kaiser–Meyer–Olkin (KMO) and Bartlett's tests indicated that these variables can provide a reasonable basis for factor analysis (KMO = 0.729 > 0.7 , $P < 0.001$).

First, the data were standardized and then used for dimensionality reduction (Table 3). We extracted $F1$

(eigenvalues = 3.590 > 1 , cumulative = 89.751%). Using the coefficient score matrix and normalization processing, we reached the following conclusion:

$$F1 = 0.13V - 0.15BD + 0.14SA + 0.15MWC \quad (13)$$

Next, we performed principal component regression analysis ($R^2 = 0.683$, $P = 0.001$), expressed the linear regression equation as $K_s = 0.086 + 0.222F1$, and combined it with Equation (9) to obtain the expression of principal component linear regression.

$$K_s = 0.086 + 0.02886V - 0.0333BD + 0.03108SA + 0.0333MWC \quad (14)$$

The effect of soil macropores on the soil hydrodynamic process

Fig. 4a shows a significantly positive relationship between dK_s and soil macropore volume. The measured value of K_s showed a marked increase with volume, while the extent of variation in its estimated value K'_s in homogeneous soil was small, and there is no significant relationship between and soil macropore volume statistically (Fig. 4b).

The estimated values of saturated hydraulic conductivity (K'_s), initial water content and saturated water content are shown in Table 4. The difference between measured and estimated values of infiltration time (dt) was used to explain how soil macropores affect soil water movement in the vertical direction. A significant positive correlation between dt and the soil macropore volume can be

Table 2: Soil properties of different soil columns

Plot number	Soil depth (cm)	Soil BD	Soil water content	MWC	SA
A1	0–10	1.470	0.233	0.312	0.414
	10–20	1.442	0.238	0.319	0.415
	20–40	1.482	0.228	0.305	0.409
	40–60	1.585	0.200	0.256	0.367
A2	0–10	1.362	0.239	0.361	0.448
	10–20	1.403	0.213	0.330	0.425
	20–40	1.445	0.162	0.317	0.430
	40–60	1.408	0.253	0.345	0.438
A3	0–10	0.979	0.236	0.563	0.516
	10–20	1.267	0.215	0.388	0.456
	20–40	1.385	0.240	0.353	0.445
	40–60	1.302	0.223	0.374	0.450

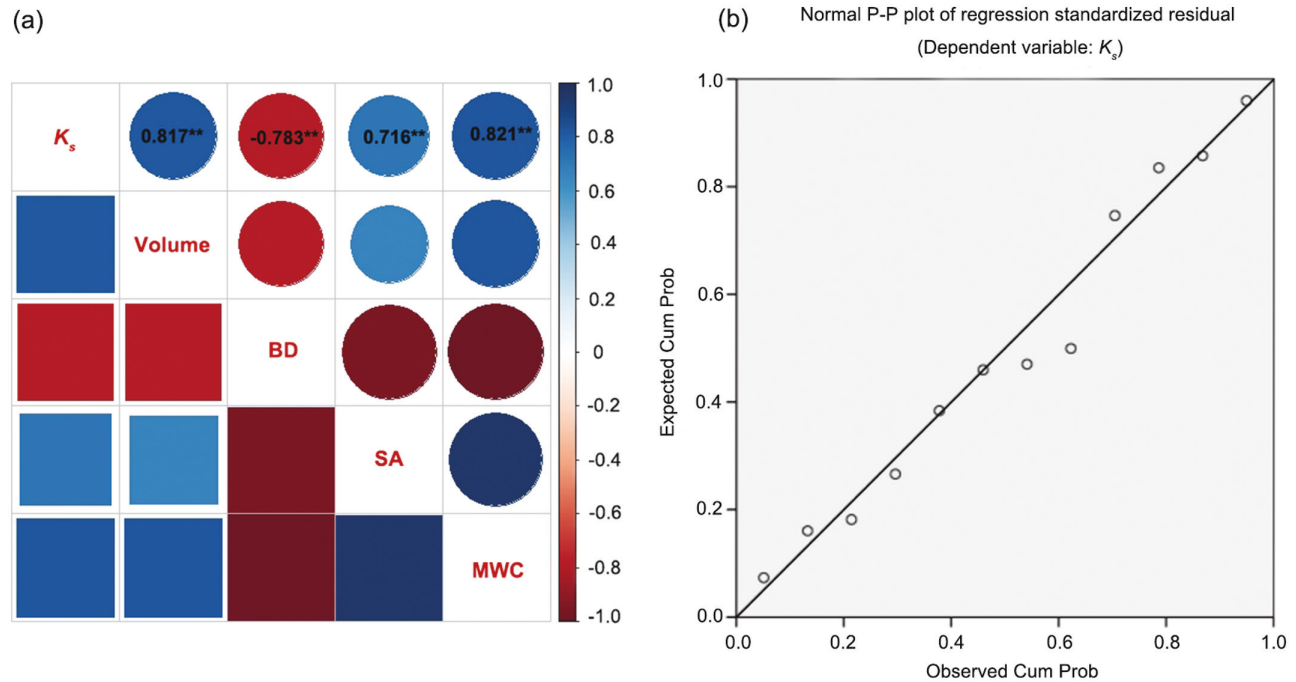


Figure 3: Results of (a) correlation analysis and (b) residual analysis. **Correlation is significant at the 0.01 level (two-tailed). *Correlation is significant at the 0.05 level (two-tailed).

Table 3: Total variance explained

Component	Initial eigenvalues			Extraction sums of squared loadings		
	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %
1	3.590	89.751	89.751	3.590	89.751	89.751
2	0.383	9.577	99.328			
3	0.020	0.498	99.826			
4	0.007	0.174	100			

observed from Fig. 5, and the correlation coefficient was 0.822 ($P = 0.007 < 0.01$). The fitting equation between Δt and V is $dt = 17.665 \log(V) + 20.704$ ($R^2 = 0.6522$). The image of the function showed that the existence of soil macropores accelerates the process of water movement, and the effect becomes decreasingly obvious with the increase in the soil macropore volume.

DISCUSSION

Soil macropore and plant types

The macropore volume was distinctly different in different plant types at different soil depths. The K_s values of the top soil layer were higher, this may be that the density of roots is richer in the top layer, and the organic layer generally appears on the soil

surface. However, in A2, the total macropore volume is much smaller than the values in A1 and A3, which may be because of the spatial distribution pattern of the plant root system and plant density. *Robinia pseudoacacia* is a kind of tree, so A3 had the largest soil macropore volume, and the K_s was also larger. As for the differences between A1 and A2, we believe they could be related to the density of vegetation (30/m² in A1, 0.4/m² in A2).

Root architecture varies greatly depending on the vegetation types, its features can determine the pathway and position of roots in soil space, the whole growth process of the plant including root development and subsequent root decay is closely associated with the macropore distribution and more continuous bio-pores in the soil (Hodge and Doussan 2009; Stewart *et al.* 1999). In addition,

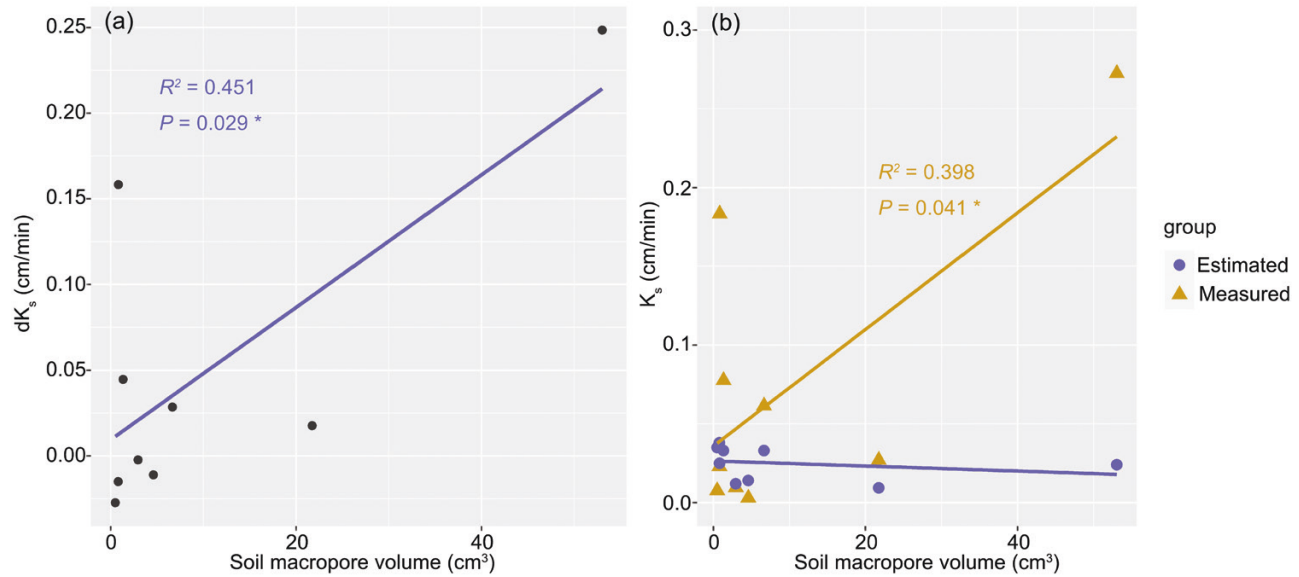


Figure 4: Relationship among soil macropore volume, dK_s , K_s and $k'_s K'_s$. (a) Regression analysis for the relationship between soil macropore volume and dK_s ; (b) Regression analysis for the relationship between soil macropore volume and measured value K_s , and the relationship between soil macropore volume and estimated value k'_s .

Table 4: Information about water parameters and the soil macropore volume

Plot number	Soil depth (cm)	K_s (cm/min)	K'_s (cm/min)	dK	θ_s	θ_i	t (min)	t_s (min)	dt (min)	V (cm ³)
A11	0–20	0.027	0.009	0.018	0.316	0.235	16.00	88.70	72.70	21.73
A12	20–40	0.003	0.014	−0.011	0.305	0.228	17.50	57.60	40.10	4.59
A13	40–60	0.010	0.012	−0.002	0.256	0.200	8.50	48.80	40.30	2.95
A21	0–20	0.062	0.033	0.029	0.345	0.226	13.50	37.70	24.20	6.65
A22	20–40	0.183	0.025	0.158	0.317	0.162	6.30	64.30	58.00	0.83
A23	40–60	0.008	0.035	−0.027	0.345	0.253	40.60	27.50	−13.10	0.49
A31	0–20	0.273	0.024	0.249	0.475	0.226	0.75	108.00	107.30	53.02
A32	20–40	0.078	0.033	0.045	0.353	0.240	4.00	35.60	31.60	1.33
A33	40–60	0.023	0.038	−0.015	0.374	0.223	27	41.4	14.4	0.8

the interactions among roots, root exudation-mediated plant–microbe and saprotrophic organisms also can enrich the soil macropores and functional characteristics (Hallett *et al.* 2009; Vishwakarma *et al.* 2017).

Interaction between K_s and soil physical properties

Recent studies have focused on revealing the coupling mechanisms between HC structurally and functionally (Lee *et al.* 2023; Zhang *et al.* 2021), here, we can see soil static physical structure characteristics as a pathway basis for soil water movement in the

vertical direction. Investigating the relationship between soil hydraulic parameters and soil properties using mathematical statistical methods can effectively explain the impact of the soil structure on ecological processes and functions. Spatial heterogeneity is a basic property of soil, even at a distance of a few centimeters, there may be a wide variation in soil hydraulic properties (Mallarino 1996).

K_s is dominated by the comprehensive effects of multiple soil physical factors rather than by merely the total porosity or soil BD (Nakano and Miyazaki 2005). In our study, K_s varied greatly in different plant communities and soil depths, which resulted

in spatial heterogeneity. K_s was negatively correlated with BD and positively correlated with macropores volume, SA and MWC, which is consistent with the results of most relevant studies (Hu *et al.* 2020; Liu *et al.* 2021b; Osunbitan *et al.* 2005). As for our multiple linear regression model, we performed the Durbin–Watson test to judge whether there was autocorrelation in the residuals of regression analysis. We obtained a value of 2.256, and we did not see an obvious autocorrelation trend in the result. In the normal distribution, the data were distributed along diagonal directions, and the regression model satisfied the normal hypothesis (Fig. 4b). Therefore, we considered that this model can combine the parameters of K_s and soil properties effectively and show good stability (Table 5).

From the perspective of soil macropores, the relationship between soil volume and hydraulic conductivity was analyzed separately, and the result showed that there is a significant positive correlation between soil macropore volume and measured hydraulic conductivity (Fig. 4), meanwhile, the

macropore can enlarge the difference between K_s and K'_s , indicating that the presence of soil macropores can promote the vertical movement of water flow. For compact soil without macropores, water does not infiltrate at once because of poor permeability. In the coastal wetlands under study, the correlation between salt-affected silt soil and K_s is negative (Mao *et al.* 2016). In A1, K_s and K'_s were smaller than in A2 and A3, which could be because the silt occupied a larger proportion in this community.

Interaction between soil structure and soil HC

The existence of soil macropores has a certain influence on water infiltration compared with homogeneous soil without macropores. Soil macropores can change soil permeability and accelerate water movement. In our study, we mainly focused on volume, but the soil structure 3D parameters, such as tortuosity, density and tilt angle density, also affect soil water movement. Previous studies have shown that the larger the macropore tortuosity, the more curved the macropore (Luo *et al.* 2010a). Curved macropores can

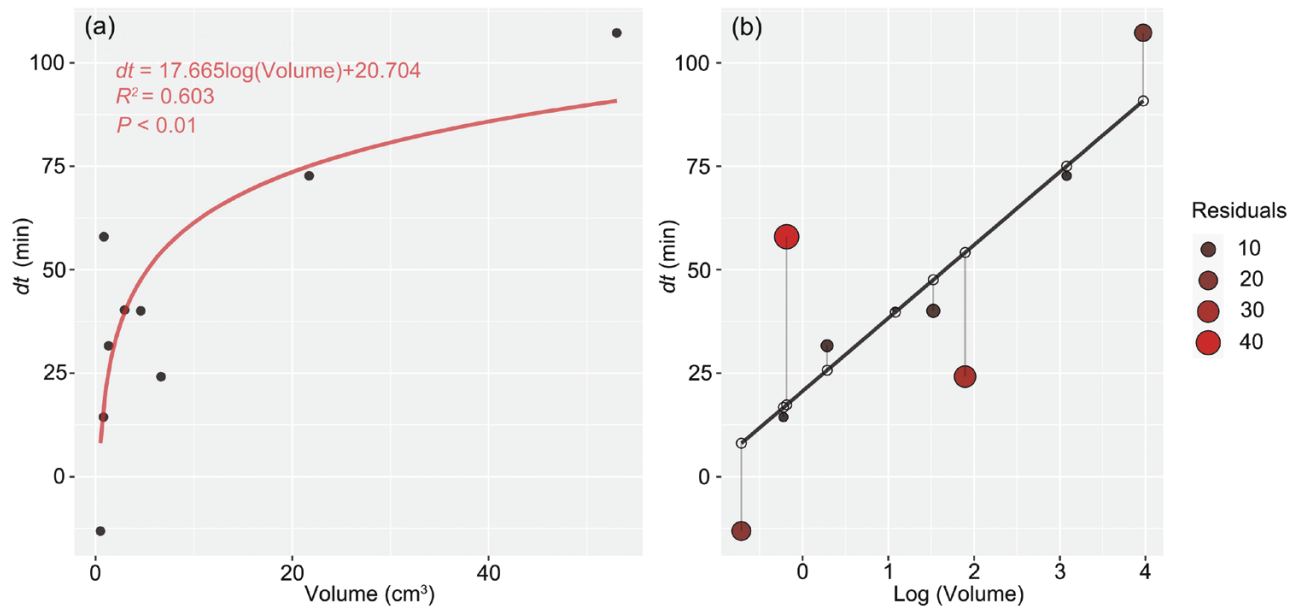


Figure 5: Relationship between soil macropore volume and dt . (a) Fitting curve equation between soil macropore volume and dt . (b) Residual analysis for the fitting curve equation; the size of circle represents the value of the residual, and the distance to the straight line represents the degree of deviation.

Table 5: Model summary^a

Model	R	R ²	Adjusted R ²	Standard error of the estimate	Durbin–Watson test
1	0.827 ^b	0.683	0.652	0.08555	2.256

^aDependent variable: K_s ; ^bPredictors: (constant), $F1$.

block and slow down the process of water infiltration, which is not conducive to rapid infiltration of water. This may explain why the estimated time was smaller than the actual time in our study. One thing we should focus on is that only macropores connected to the surface are accessible to the infiltrating water (Bastardie *et al.* 2003), so the interconnectedness of the soil macropore system may be more important than the volume index. In our study, the length of the soil column used was limited, which might be inconsistent with the actual situation.

Structure and function have always been inseparable, whether in biological or in nonbiological fields. The function of a specific system would be better understood based on the comprehensive analysis of its physical structure (Rabot *et al.* 2018). The better mobility of soil water is the condition to promote small-scale vertical HC, so in this article, we emphasized the contribution of soil structure to HC. We mainly focused on the vertical flow. Due to the limitation of the soil column wall, the horizontal flow induced by preferential pathways in the organic layer of the soil and the water-repellent layer occurring between the organic layer and the mineral layer were ignored.

As for coastal wetlands, the presence of the soil macropore structure, such as crab burrows and plant roots, can greatly increase tidally driven water exchange and improve the intensity and duration of soil respiration (Xiao *et al.* 2019; Xie *et al.* 2022), which would be a great significance to maintain ecosystem function. We found that increasing soil microporosity improved the convenience of water movement, which would enhance the HC of the region and ameliorate the function of saline soils. Vegetation restoration is also helpful to improve the surface-saturated hydraulic conductivity (Boldrin *et al.* 2022). Therefore, small-scale HC can be improved by plant growth and increasing soil macropores in the YRD. For practical work, it is necessary to input more effort into replanting in degradation areas to improve soil structure and promote small-scale HC, and the plant species should be taken into consideration.

The improvement of our understanding of fundamental processes in natural sciences is more dependent on the advances with technological and methodological progress (Vereecken *et al.* 2015), and the accurate quantification of the soil macropore structure will help us grasp the natural law better. The differences between the distribution of soil macropores may be caused by the root system and its connectivity parameters, while root growth is a

dynamic process, which should be considered in further studies.

CONCLUSIONS

Soil macropores are beneficial to the increase of hydrological conductivity of clayey saline soil in the YRD, the soil structure changes with the growth of plant roots, leading to changes in HC inside wetlands at a small scale. Depending on the urgent need to restore wetlands, it is necessary to focus on the soil water movement, promoting the understanding of the connotations of HC from microcosmic perspectives.

In our study, we obtained the structural characteristics of soil macropores accurately by X-ray technology and found spatial heterogeneity in the soil macropores and K_s . K_s is negatively correlated with BD and positively correlated with volume, SA and MWC. We also evaluated the effect of the soil macropore volume on soil water movement, and found abundant soil macropores could improve vertical hydrological function connectivity at an extent. This result can provide a theoretical basis for vertical hydrological function connectivity and guide wetland conservation and restoration.

Funding

This research was supported by the Young Scientists Fund of the National Natural Science Foundation of China (42307573) and Fundamental Research Funds for the Central Universities in China (BLX202250). We extend thanks to many participants who helped us to do this experiment.

Authors' Contributions

Lumeng Xie: conceptualization (lead), investigation and writing—original draft (lead); Jiakai Liu: writing—original draft (supporting); Yi Li: investigation (supporting) and data curation (lead); Peisheng Huang: review and editing (equal) and methodology (lead); Matt Hipsey: conceptualization (supporting) and project administration; Mingxiang Zhang: review and editing (equal); Zhenming Zhang: conceptualization (supporting), funding acquisition and project administration.

Conflict of interest statement. The authors declare that they have no conflict of interest.

REFERENCES

- Altieri AH, Cui BS (2015) Herbivory drives zonation of stress-tolerant marsh plants. *Ecology* **96**:1318–1328.
- Bai J, Yu Z, Yu L, *et al.* (2019) In-situ organic phosphorus mineralization in sediments in coastal wetlands with

- different flooding periods in the Yellow River Delta, China. *Sci Total Environ* **682**:417–425. <https://doi.org/10.1016/j.scitotenv.2019.05.176>
- Banwart SA, Nikolaidis NP, Zhu YG, *et al.* (2018) Soil functions: connecting earth's critical zone. *Annu Rev Earth Planet Sci* **47**:333–359. <https://doi.org/10.1146/annurev-earth-063016-020544>
- Bastardie F, Capowicz Y, Dreuz J, *et al.* (2003) X-ray tomographic and hydraulic characterization of burrowing by three earthworm species in repacked soil cores. *Appl Soil Ecol* **24**:3–16.
- Beven K, Germann P (1982) Macropores and water flow in soils. *Water Resour Res* **18**:1311–1325. <https://doi.org/10.1029/wr018i005p01311>
- Boldrin D, Knappett J, Leung A, *et al.* (2022) Modifying soil properties with herbaceous plants for natural flood risk-reduction. *Ecol Eng* **180**:106668.
- Bonell M, Purandara BK, Venkatesh B, *et al.* (2010) The impact of forest use and reforestation on soil hydraulic conductivity in the Western Ghats of India: implications for surface and sub-surface hydrology. *J Hydrol* **391**:47–62. <https://doi.org/10.1016/j.jhydrol.2010.07.004>
- Bottinelli N, Zhou H, Capowicz Y, *et al.* (2017) Earthworm burrowing activity of two non-Lumbricidae earthworm species incubated in soils with contrasting organic carbon content (Vertisol vs. Ultisol). *Biol Fertil Soils* **54**:173. <https://doi.org/10.1007/s00374-017-1240-y>
- Conte P, Ferro V (2019) Standardizing the use of fast field cycling NMR relaxometry for measuring hydrological connectivity inside a soil. *Magn Reson Chem* **58**:41–50. <https://doi.org/10.1002/mrc.4907>
- Dai L, Zhang Y, Liu Y, *et al.* (2020a) Assessing hydrological connectivity of wetlands by dye-tracing experiment. *Ecol Indic* **119**:106840. <https://doi.org/10.1016/j.ecolind.2020.106840>
- Dai L, Zhang Y, Liu Y, *et al.* (2020b) Thick roots and less microaggregates improve hydrological connectivity. *Chemosphere* **266**:129008.
- Guo B, Liu Y, Fan J, *et al.* (2023) The salinization process and its response to the combined processes of climate change–human activity in the Yellow River Delta between 1984 and 2022. *Catena* **231**:107301. <https://doi.org/10.1016/j.catena.2023.107301>
- Hallett PD, Feeney DS, Bengough AG, *et al.* (2009) Disentangling the impact of AM fungi versus roots on soil structure and water transport. *Plant Soil* **314**:183–196. <https://doi.org/10.1007/s11104-008-9717-y>
- Hodge B, Doussan MC (2009) Plant root growth, architecture and function. *Plant Soil* **321**:153–187.
- Horn R, Mordhorst A, Fleige H, *et al.* (2020) Soil type and land use effects on tensorial properties of saturated hydraulic conductivity in Northern Germany. *Eur J Soil Sci* **71**:179–189. <https://doi.org/10.1111/ejss.12864>
- Hu X, Li X-Y, Li Z-C, *et al.* (2020) Linking 3-D soil macropores and root architecture to near saturated hydraulic conductivity of typical meadow soil types in the Qinghai Lake Watershed, northeastern Qinghai–Tibet Plateau. *Catena* **185**:104287. <https://doi.org/10.1016/j.catena.2019.104287>
- Hu X, Li X-Y, Wang P, *et al.* (2019) Influence of enclosure on CT-measured soil macropores and root architecture in a shrub-encroached grassland in northern China. *Soil Tillage Res* **187**:21–30. <https://doi.org/10.1016/j.still.2018.10.020>
- Huang L, Shao M (2019) Advances and perspectives on soil water research in China's Loess Plateau. *Earth-Sci Rev* **199**:102962. <https://doi.org/10.1016/j.earscirev.2019.102962>
- Ilek A, Kucza J, Witek W (2019) Using undisturbed soil samples to study how rock fragments and soil macropores affect the hydraulic conductivity of forest stony soils: some methodological aspects. *J Hydrol* **570**:132–140. <https://doi.org/10.1016/j.jhydrol.2018.12.067>
- Jahanishakib F, Salmanmahiny A, Mirkarimi SH, *et al.* (2021) Hydrological connectivity assessment of landscape ecological network to mitigate development impacts. *J Environ Manage* **296**:113169. <https://doi.org/10.1016/j.jenvman.2021.113169>
- Jarvis N, Moeys J, Hollis J, *et al.* (2009) A conceptual model of soil susceptibility to macropore flow. *Vadose Zone J* **8**:902–910.
- Kochieru M, Lamorski K, Feiza V, *et al.* (2020) Quantification of the relationship between root parameters and soil macropore parameters under different land use systems in Retisol. *Int Agrophys* **34**:301–308. <https://doi.org/10.31545/intagr/123266>
- Lee E, Epstein JM, Cohen MJ (2023) Patterns of wetland hydrologic connectivity across coastal-plain wetlandscapes. *Water Resour Res* **59**:e2023WR034553.
- Liu J, Engel BA, Zhang G, *et al.* (2020) Hydrological connectivity: one of the driving factors of plant communities in the Yellow River Delta. *Ecol Indic* **112**:106150. <https://doi.org/10.1016/j.ecolind.2020.106150>
- Liu Y, Zhang Y, Xie L, *et al.* (2021) Effect of soil characteristics on preferential flow of *Phragmites australis* community in Yellow River delta. *Ecol Indic* **125**:107486.
- Liu B, Fan H, Jiang Y, *et al.* (2023) Evaluation of soil macro-aggregate characteristics in response to soil macropore characteristics investigated by X-ray computed tomography under freeze-thaw effects. *Soil Tillage Res* **225**:105559. <https://doi.org/10.1016/j.still.2022.105559>
- Liu J, Liu Y, Xie L, *et al.* (2021a) A threshold-like effect on the interaction between hydrological connectivity and dominant plant population in tidal marsh wetlands. *Land Degrad Dev* **32**:2922–2935. <https://doi.org/10.1002/ldr.3913>
- Liu S, Wang Y, An Z, *et al.* (2021b) Watershed spatial heterogeneity of soil saturated hydraulic conductivity as affected by landscape unit in the critical zone. *Catena* **203**:105322. <https://doi.org/10.1016/j.catena.2021.105322>
- Lu J, Zhang Q, Werner AD, *et al.* (2020) Root-induced changes of soil hydraulic properties – A review. *J Hydrol* **589**:125203.
- Luo L, Lin H, Li S (2010a) Quantification of 3-D soil macropore networks in different soil types and land uses using computed tomography. *J Hydrol* **393**:53–64. <https://doi.org/10.1016/j.jhydrol.2010.03.031>
- Luo L, Lin H, Schmidt J (2010b) Quantitative relationships between soil macropore characteristics and preferential flow and transport. *Soil Sci Soc Am J* **74**:1929–1937. <https://doi.org/10.2136/sssaj2010.0062>

- Mahmood WI, Mohammed AS (2020) Hydraulic conductivity, grain size distribution (GSD) and cement injectability limits predicted of sandy soils using Vipulanandan models. *Geotech Geol Eng* **38**:2139–2158. <https://doi.org/10.1007/s10706-019-01153-z>
- Mallarino AP (1996) Spatial variability patterns of phosphorus and potassium in no-tilled soils for two sampling scales. *Soil Sci Soc Am* **60**:1473–1481.
- Mao W, Kang S, Wan Y, *et al.* (2016) Yellow river sediment as a soil amendment for amelioration of saline land in the Yellow River Delta. *Land Degrad Dev* **27**:1595–1602. <https://doi.org/10.1002/ldr.2323>
- Najm M, Jabro JD, Iversen WM, *et al.* (2010) New method for the characterization of three-dimensional preferential flow paths in the field. *Water Resour Res* **46**:W02503.
- Nakano K, Miyazaki T (2005) Predicting the saturated hydraulic conductivity of compacted subsoils using the non-similar media concept. *Soil Tillage Res* **84**:145–153. <https://doi.org/10.1016/j.still.2004.11.010>
- Osunbitan JA, Oyedele DJ, Adekalu KO (2005) Tillage effects on bulk density, hydraulic conductivity and strength of a loamy sand soil in southwestern Nigeria. *Soil Tillage Res* **82**:57–64. <https://doi.org/10.1016/j.still.2004.05.007>
- Rabot E, Wiesmeier M, Schlüter S, *et al.* (2018) Soil structure as an indicator of soil functions: a review. *Geoderma* **314**:122–137. <https://doi.org/10.1016/j.geoderma.2017.11.009>
- Stewart JB, Moran CJ, Wood JT (1999) Macropore sheath: quantification of plant root and soil macropore association. *Plant Soil* **211**:59–67.
- Vereecken H, Huisman JA, Franssen HH, *et al.* (2015) Soil hydrology: recent methodological advances, challenges, and perspectives. *Water Resour Res* **51**:2616–2633.
- Vishwakarma K, Mishra M, Jain S, *et al.* (2017). Exploring the role of plant-microbe interactions in improving soil structure and function through root exudation: a key to sustainable agriculture. *Plant-Microbe Interactions in Agro-Ecological Perspectives: Volume 1: Fundamental Mechanisms, Methods and Functions*. Singapore: Springer, 467–487.
- Wu L, Zhang M, Xie X, *et al.* (2023) Effect of wetland hydrological connectivity on runoff based on a basin comparison. *J Hydrol* **619**:129348. <https://doi.org/10.1016/j.jhydrol.2023.129348>
- Xiao K, Wilson AM, Li H, *et al.* (2019) Crab burrows as preferential flow conduits for groundwater flow and transport in salt marshes: a modeling study. *Adv Water Resour* **132**:103408. <https://doi.org/10.1016/j.advwatres.2019.103408>
- Xie L, Wang Y, Zhao S, *et al.* (2022) Sizes of crab burrows regulate water-salt transport of tidal marsh wetlands. *Mar Environ Res* **179**:105691. <https://doi.org/10.1016/j.marenvres.2022.105691>
- Xingyuan Z, Fawen L, Yong Z (2023) Impact of changes in river network structure on hydrological connectivity of watersheds. *Ecol Indic* **146**:109848. <https://doi.org/10.1016/j.ecolind.2022.109848>
- Ying Z, Zhang C, Wang Z, *et al.* (2016) Vegetation dynamics and its driving forces from climate change and human activities in the three-river source region, China from 1982 to 2012. *Sci Total Environ* **56**:210–220.
- Zhang Y, Huang C, Zhang W, *et al.* (2021) The concept, approach, and future research of hydrological connectivity and its assessment at multiscales. *Environ Sci Pollut Res Int* **28**:52724–52743. <https://doi.org/10.1007/s11356-021-16148-8>
- Zhang Z, Liu K, Zhou H, *et al.* (2019) Linking saturated hydraulic conductivity and air permeability to the characteristics of biopores derived from X-ray computed tomography. *J Hydrol* **571**:1–10. <https://doi.org/10.1016/j.jhydrol.2019.01.041>
- Zhang Y, Niu J, Zhang M, *et al.* (2016) Interaction between plant roots and soil water flow in response to preferential flow paths in northern China. *Land Degrad Dev* **28**: 648–663. <https://doi.org/10.1002/ldr.2592>
- Zheng P, Jiang X, Shu F, *et al.* (2022) Loss of lateral hydrological connectivity impacts multiple facets of molluscan biodiversity in floodplain lakes. *J Environ Manage* **320**:115885. <https://doi.org/10.1016/j.jenvman.2022.115885>
- Zhou S, Wang C, Li Y, *et al.* (2023) Study on spatio-temporal variation and hydrological connectivity of tidal creek evolution in Yancheng coastal wetlands. *Environ Sci Pollut Res Int* **30**:37143–37156. <https://doi.org/10.1007/s11356-022-24871-z>